

ISO 8583 Specification For Airtel Money

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CHAPTER 1

Design and Integration

1.1 Communication

Connection between *3PFI* and Airtel Money will be socket-based following the server-client architecture. Airtel Money will be the server while *3PFI* the client.

As TCP is a stream-oriented protocol, the first four bytes of all messages transmitted will indicate the length of the message (excluding the length indicator). For example if the message length is 71 bytes then 0071 should be prefixed to the TCP/IP stream.

1.2 Security

Secure pins will be processed from *3PFI* to Airtel Money with the following parameters:
o Shared (2-part component) single length static pin working key to be configured at both *3PFI* and Airtel Money

- Pin block format: ISO-0 (ANSI X9.8, VISA-1, ECI)
- A pin verification key, PVK, to be used by Airtel Money to verify secure pins against stored pin offsets

3PFI and Airtel Money connectivity will be on a private lease line and therefore, no extra data encapsulation and data security will be required. Access-lists will be defined for socket security.

1.3 Message Type

The following message identifiers will be supported between *3PFI* and Airtel Money:

- 0200 online financial transaction request – bidirectional
- 0210 online financial transaction response – bidirectional
- 0420 reversal advice – bidirectional
- 0421 repeat reversal advice – bidirectional
- 0430 reversal advice - bidirectional
- 0800 network management request
- 0810 network management response

1.4 Network Management Processing

Network management functions between *3PFI* and Airtel Money will be implemented using the ISO8583 (93) 08xx series network management messages.

- 0800 – network management request
- 0810 – network management request response

Sign on request and response

Sign on message shall be implemented to be originated by *3PFI* to Airtel Money, for Airtel Money advise on readiness to start processing transaction requests from *3PFI* upon successful establishment of connectivity.

Echo Test (polling)

The purpose of the echo test is for either *3PFI* or Airtel Money to determine the status of the other party. *3PFI* will implement a frequency of 5 minutes (configurable).

1.5 Message Timeout

The initiator of a transaction request will establish the amount of time it will wait for a response time-out. The time-out period will be set to 30 seconds (configurable)

Initiator of a transaction request will immediately initiate a reversal and continuously send repeat reversals until the reversal is acknowledged. The time-out period will apply to both a transaction request and a reversal advice

1.6 Message Matching

The following fields will be used to match messages transmitted between *3PFI* and Airtel Money:

1. Message Type Identifier
2. Account number
3. Time, Local Transaction
4. Date, Local Transaction
5. System Trace Audit Number

CHAPTER 2

Message Format

2.1 Message Structure

The messages transmitted between the *3PFI* and Airtel Money will be constructed with the basic message structure prescribed below. Messages are constructed in the sequence message.

Message type identifier	1 or 2 hex bitmaps	Data elements
-------------------------	--------------------	---------------

2.2 Message Fields Mapping

Notations:

- 'M' – mandatory, must be populated with valid data,
- 'C' – conditional, required if certain conditions are met
- 'O' – optional, may be included or not
- 'N' – Never required

Field No	Name	0200	0210	0420 0421	0430	0800	0810
-	Length	M	M	M	M	M	M
-	Message type	M	M	M	M	M	M
-	Bitmap	M	M	M	M	M	M
2	Pan	M	O	C	O	N	N
3	Processing code	M	M	M	O	N	N
4	Amount Transaction	M	M	M	O	N	N
7	Transmission Date and time	M	M	M	O	M	M
11	System trace Audit number	M	M	M	O	M	M
12	Local Time	M	M	M	O	M	M
13	Local Date	M	M	M	O	M	M
18	Merchant Type	O	O	O	O	N	N
30	Commission	M	M	M	O	N	N
32	Acquiring Institution bin	M	O	M	O	N	N
37	Retrieval Reference number	M	M	M	O	N	N
38	Authorization code	C	C	O	O	N	N
39	Response code	N	M	C	M	N	M
41	Terminal ID	M	O	M	O	N	N
42	Card acceptor ID code	M	O	M	O	N	N
43	Card acceptor name and loc	M	O	M	O	N	N
48	Additional response data	N	C	N	O	N	N
49	Currency code	M	O	M	O	N	N

52	PIN Data	C	N	N	O	N	N
54	Additional amounts	C	C	C	O	N	N
59	Echo data	O	M	M	M	N	N
70	Network management info	N	N	N	O	M	M
90	Original data elements	N	N	M	O	N	N
95	Replacement amounts	N	N	M	C	N	N
102	Account to debit	M	C	M	C	N	N
103	Account to credit	M	C	M	C	N	N

Fields Definition

3.1 Data Types

Please note the following default values for the more common data types.

A - Alphabetic character, A through Z and a through z

n - Numeric digits, 0 through 9

ans - alphabetic, numeric and special characters

DD - Date, 01 through to 31

MM - Month, 01 through to 12

YY - Year, 00 through to 99 no apparent allowance for the turn of the century.

Hh - Hour, 00 through 23 mm - Minute, 00 through 59 ss - Second, 00 through 59

LL - length of variable length data element that follows, 01 through 99

LLL - length of variable length data element that follows, 01 through 999

3 - Fixed length of 3 characters ..11 - Variable length up to 11

b - Binary representation of data z - track 2 code

VAR - Variable Length Data Field

x - C-Credit or D-Debit normally prefixes a numeric amount time - time format

date - date format

Note – all fixed length “n” elements are presumed to be right justified with leading zeroes, except for those where examples show left justified with trailing spaces. All other fixed length data elements are left justified with trailing spaces.

3.2 Field Definition

1. Length

Format: n4

Description: this is length of the message. I.e. the number of characters contained in the message being sent e.g. 0199

2. Message Type

Format: n4

Description: this is used to specify the message type if e.g. 0200 for a Bank registration

3. Bitmap

Format: n32

Description: e.g. E23000040021800000000000040000001

4. Field 2: Primary Account Number (PAN)

Format: n...19, LLVAR

Description: This is a number identifying the cardholder and the card issuer. Typically, this number is embossed on the front of the card and encoded on Track 2 of the magnetic stripe. This can also be the MSISDN of the customer e.g. 255689884714

5. Field 3: Processing Code

Format: n6

Description: The customer transaction type and the account types, if any, affected by the transaction; this is a fixed length field consisting of 3 data elements:

Transaction type code:

Digits 1 and 2	Description
0	Balance Inquiry

01	Mini statement
02	Bank to Airtel Money
03	Airtel Money to Bank
04	Bank to Bank
05	Registration of Airtel Money Subscriber
06	Bank account Registration
07	Bank to Airtel Money (Bank Originated)

From account type

Digits 3 and 4	Description
00	Default Account
10	Savings
20	Cheque

To account type

Digits 5 and 6	Description
00	Default Account
10	Savings
20	Cheque

E.g. for a bank registration 060000

6. Field 4: Amount Transaction

Format: n12

Description: The funds requested by the cardholder in the local currency of the acquirer or source location of the transaction exclusive of transaction fees. Values are expressed in the minor denomination (e.g. cents). I.e. a fund request for 100.00 is written as 000000010000

7. Field 7: Transmission Date and Time

Format: n10, MMDDhhmmss

Description: The date and time, expressed in Coordinated Universal Time (UTC), when this message is sent by the message initiator. E.g. 0919050516 for a transmission done at '9/19/2013 5:05:16 AM'

8. Field 11: System Trace Audit Number

Format: n6

Description: A number assigned by a transaction originator to assist in identifying a transaction uniquely. E.g. 161921

9. Field 12: Time, Local Transaction

Format: n6, hhmmss

Description: The local time at which the transaction takes place at the card acceptor location in authorization and financial message.

10. Field 13: Date, Local Transaction

Format: n4, MMDD

Description: The local date at which the transaction takes place at the card acceptor location in authorization and financial messages

11. Field 18: Merchant Type

Format: n4

Description: The classification of the merchant's type of business product or service, codes to be developed within each country

12. Field 30: Commission

Format: ans9

Description: this field should include the fixed data of **D00000000**.

13. Field 32: Acquiring Institution Id Code

Format: n...11n, LLVAR

Description: A code identifying the financial institution acting as the acquirer of this customer transaction; the acquirer is the member or system user that signed the merchant, installed the ATM or dispensed cash. E.g. 639485

14. Field 37: Retrieval Reference Number

Format: an12

Description: A reference number supplied by the system retaining the original source information and used to assist in locating that information or a copy thereof. E.g. 000008033167

15. Field 38: Authorization Id Response

Format: anp6

Description: A code assigned by the authorizing institution indicating approval

16. Field 39: Response Code

Format: an2

Description: A code that defines the disposition of a transaction

Code	Description
00	Successful
01	Failed

17. Field 41: Card Acceptor Terminal Id

Format: ans8

Description: A unique code identifying a terminal at the card acceptor location

18. Field 42: Card Acceptor Id Code

Format: ans15

Description: A code identifying the card acceptor (typically a merchant)

19. Field 43: Card Acceptor Name and Location

Format: ans40

Description: The name and location of the card acceptor (such as a merchant or an ATM)

20. Field 48: Additional Data

Format: ans....9999, LLLLVAR

Description: Used to provide linked account or mini-statement information for a linked account inquiry or a mini-statement inquiry. It can also be used to provide more data about the transaction, for example. bank account details. The data in this field could contain more than one data element separated by a delimiter, most commonly a comma (,). **E.g.** FIRST NAME: Moses, SECOND NAME:, LAST NAME: Mwangi, NATIONALITY:KENYA, DATE OF BIRTH:05-June-1983, ID NUMBER:°aE}□²+U.\! HDÂ, ACCOUNT STATUS:ACTIVE

21. Field 49: Currency Code**Format:** n3**Description:** The local currency of the acquirer or source location of the transaction e.g. 404 to represent KES. More codes on <http://www.nationsonline.org/oneworld/currencies.htm>**22. Field 52:** Pin Data**Format:** b8 (Hex 16)**Description:** The PIN data field contains the PIN (a number assigned to a cardholder intended to uniquely identify that cardholder) of the cardholder formatted into a 64-bit block and encrypted with a DES key.**23. Field 54:** Additional Amounts**Format:** an...120, LLLVAR**Description:** Information on up to 6 amounts and related account data for which specific data elements have not been defined, each amount is a fixed length field consisting of 5 data elements: e.g.

- Account Type (positions 1 - 2)
- Amount Type (positions 3 - 4) : 01=ledger bal; 02=available bal
- Currency code (positions 5 - 7)
- Amount sign (position 8) - "C" or "D"
- Amount (position 9 - 20)

24. Field 59: Echo Data**Format:** ans...225, LLLVAR**Description:** Contains data from the originator of the message that shall be returned unaltered in the response message e.g. 10736**25. Field 70:** Network Management Information Code**Format:** n3**Description:** The code that defines the type of network management needed as follows: 001=Sign On Request 301=Echo Test**26. Field 90:** Original Data Elements**Format:** n42**Description:** The data elements contained in the original message intended for transaction matching (e.g. to identify a transaction for correction or reversal). It is a fixed length field consisting of 5 data elements:

- Original message type (positions 1 - 4) - the message type identifier of the original message of the transaction being reversed. E.g. 0200 –financial transaction
- Original systems trace audit number (positions 5 - 10) - the System Trace Audit Number of the original message.
- Original transmission date and time (positions 11 - 20) – the transmission date and time of the original message
- Original acquirer institution ID code (position 21 - 31) – the Acquirer Institution Id Code of the original message (right justified with leading zeroes)
- Original forwarding institution ID code (position 32 - 42) – the Forwarding Institution Id Code of the original message (right justified with leading zeroes)

e.g. 02008269690925075944000006275590000000000

27. Field 95: Replacement Amounts**Format:** an42**Description:** The corrected amounts of a transaction in a partial or full reversal (or the final amounts for the transaction), it is a fixed length field consisting of 4 data elements:

- Actual amount, transaction (positions 1 - 12) - the corrected, actual amount of the customer's transaction, in the currency of the transaction
- Actual amount, settlement (positions 13 - 24) - the corrected, actual amount of the customer's transaction, in the settlement currency
- Actual amount, transaction fee (positions 25 - 33) - the corrected, actual amount of the fee (in format x + n8) for this customer transaction, in the currency of the transaction
- Actual amount, settlement fee (positions 34 - 42) - the corrected, actual amount of the fee (in format x + n8) for this customer transaction, in the settlement currency

28. Field 102: Account Identification 1**Format:** ans...28, LLVAR**Description:** A series of digits and/or characters used to identify a specific account held by the cardholder at the card issuer and if present, shall remain unchanged for the life of the transaction. This field usually contains the description of the "from" account e.g. 01D1000099102

F48: LIMITED MOCARAN
F49: 404
F102: 0910298904845
F103: 0001400303320

Response

02061200F238000500A1800000000000060000001225478999223702000000000070000
008010826220006510826220801D00000000010NOT DEFIBANK TO ZAP 0015LIMITED
MOCARAN404130910298904845130001400303320
Length: 0206
Message type: 1210
Bitmap: F238000500A18000000000000600000012
F2: 254789992237
F3: 020000
F4: 000000700000
F7: 0801082622
F11: 000651
F12: 082622
F13: 0801
F30: D00000000
F32: 0
F39: 00
F41: NOT DEFI
F43: BANK TO ZAP
F48: 0001404C0043050593280002404C004325943296
F49: 404
F102: 0910298904845
F103: 0001400303320

4.4 Airtel Money to Bank (Mobile Originated)

Request

02061200F238000500A180000000000006000000122547363963510300000000001000
008011213460006551213460801D00000000010NOT DEFIZAP TO BANK 0015JARUPIA
IBRAHIM404130001400303320131050161280303
Length: 0206
Message type: 1200
Bitmap: F238000500A18000000000000600000012
F2: 254736396351
F3: 030000
F4: 000000010000
F7: 0801121346
F11: 000655
F12: 121346
F13: 0801
F30: D00000000
F32: 0
F41: NOT DEFI
F43: ZAP TO BANK
F48: JARUPIA IBRAHIM
F49: 404
F102: 0001400303320
F103: 1050161280303

Response

02721210F238000502A1800000000000060000001225473639635103000000000001000
008011213460006551213460801D0000000001000NOT DEFIZAP TO BANK 0079YOUR
ACCOUNT HAS BEEN SUCCESSFULLY CREDITED. YOUR CLEARED BALANCE IS -
3400841.0404130001400303320131050161280303
Length: 0272
MESSAGE TYPE: 1210
Bitmap: F238000502A18000000000000600000012
F2: 254736396351
F3: 030000
F4: 000000010000
F7: 0801121346
F11: 000655
F12: 121346
F13: 0801
F30: D00000000
F32: 0
F39: 00
F41: NOT DEFI
F43: ZAP TO BANK
F48: YOUR ACCOUNT HAS BEEN SUCCESSFULLY CREDITED. YOUR CLEARED BALANCE
IS -3400841.0
F49: 404
F102: 0001400303320
F103: 1050161280303

4.5 Balance Query

Request (sent to Bank)

01791200E238000500A180000000000004000000122547363963510000000801092201
0006520922010801D00000000010NOT DEFIBALANCE ENQUIRY 0015JARUPIA
IBRAHIM404131050161280303
Length: 0179
Message type: 1200
Bitmap: E238000500A18000000000000400000012
F2: 254736396351
F3: 000000
F7: 0801092201
F11: 000652
F12: 092201
F13: 0801
F30: D00000000
F32: 0
F41: NOT DEFI
F43: BALANCE ENQUIRY
F48: JARUPIA IBRAHIM
F49: 404
F102: 1050161280303

Response (From Bank)

01951210E238000502A180000000000004000000122547363963510000000801092201
0006520922010801D0000000001000NOT DEFIBALANCE ENQUIRY 0029YOUR CLEARED
BALANCE IS 46.70404131050161280303
Length: 0195
Message type: 1210
Bitmap: E238000502A18000000000000400000012
F2: 254736396351
F3: 000000

F7: 0801092201
F11: 000652
F12: 092201
F13: 0801
F30: D00000000
F32: 0
F39: 00
F41: NOT DEFI
F43: BALANCE ENQUIRY
F48: YOUR CLEARED BALANCE IS 46.70
F49: 404
F102: 1050161280303

4.6 Mini statement

Request

01781200E238000500A180000000000004000000122547356330570100000902181124
0011961811240902D00000000010NOT DEFIMINI STATEMENT 0014WACHIRA
DANIEL404130690192516155
Length: 0178
Message type: 1200
Bitmap: E238000500A18000000000000400000012
F2: 254735633057
F3: 010000
F7: 0902181124
F11: 001196
F12: 181124
F13: 0902
F30: D00000000
F32: 0
F41: NOT DEFI
F43: MINI STATEMENT
F48: WACHIRA DANIEL
F49: 404
F102: 0690192516155

Response (from bank)

03091210E238000502A180000000000004000000122547356330570100000902181124
0011961811240902D0000000001000NOT DEFIMINI STATEMENT 014331/08/13 CRASH
WITHDR DR 55.0
31/08/13 EAZZY-BAL-I DR 5.5
02/09/13 AIRTELM BAL DR 3.3
02/09/13 CRHQ DREPOSIT CR 90930.0
404130690192516155
Length:
Message type: 1210
Bitmap: E238000502A18000000000000400000012
F2: 254735633057
F3: 010000
F7: 0902181124
F11: 001196
F12: 181124
F13: 0902
F30: D00000000
F32: 0
F39: 00
F41: NOT DEFI

F43: MINI STATEMENT
F48: 31/08/13 CRASH WITHDR DR 55.0
31/08/13 EAZZY-BAL-I DR 5.5
02/09/13 AIRTELM BAL DR 3.3
02/09/13 CRHQ DREPOSIT CR 90930.0
31/08/13 EAZZY-BAL-I DR 5.5
02/09/13 AIRTELM BAL DR 3.3
02/09/13 CRHQ DREPOSIT CR 90930.0
F49: 404
F102: 0690192516155

4.7 Bank to Bank (Mobile Originated)

Request

02031200F238000500A1800000000000060000001225473372605304000000000632500
007251915400006021915400725D00000000010NOT DEFIBANK TO BANK 0012GICHUKI
JANE404130470192033433130120100303856
Length: 0203
Message type: 1200
Bitmap: F238000500A18000000000000600000012
F2: 254733726053
F3: 040000
F4: 000006325000
F7: 0725191540
F11: 000602
F12: 191540
F13: 0725
F30: D00000000
F32: 0
F41: NOT DEFI
F43: BANK TO BANK
F48: GICHUKI JANE
F49: 404
F102: 0470192033433
F103: 0120100303856

Response

Length:
Message Type:
Bitmap:
F2=MSISDN
F3=040000
F4=000000000012
F7=MMDDhhmmss
F11=REFERENCE ID
F12=hhmmss
F30=D00000000
F38=000000
F39=Response Code
F43=Bank to Bank
F48=Your account has been successfully debited. Your cleared balance is #####
F49=404
F54=Your account has been successfully credited. Your cleared balance is #####
102=BANK ACCOUNT NUMBER TO BE DEBITED
103=BANK ACCOUNT NUMBER TO BE CREDITED

4.8 Registration of AIRTEL MONEY Subscriber

Request

Length:
Message Type:
Bitmap:
F2=MSISDN
F3=050000
F7=MMDDhhmmss
F11=REFERENCE ID
F12=hhmmss
F30=D00000000
F43=AIRTEL MONEY Subscription
F48= First Name, Last Name, Street Address, Town,
Document Type, ID Number, Date Of Birth, Place Of Birth, Nationality
F49=404

Response

Length:
Message Type:
Bitmap:
F2=MSISDN
F3=050000
F7=MMDDhhmmss
F11=REFERENCE ID
F12=hhmmss
F30=D00000000
F38=000000
F39=Response Code
F43= AIRTEL MONEY Subscription
F48=Subscription Received, thank you.
F49=404

4.9 Bank to Airtel Money (Bank Originated)

Request from Bank

01911200F238000500A1800000000000060000001012345678902100000000000050000
6120411320411961534230705D00000000010NOT DEFIBANK TO
ZAP 0004NAME4041225473469027012254737561276

F2: 1234567890 -> '1234567890'-----> (Destination Account number)
F3: 210000 -> '210000'
F4: 000000005000 -> '50.00'
F7: 0612041132 -> '6/12/2015 4:11:32 AM'
F11: 041196 -> '041196' ----->(Bank unique ID)
F12: 153423 -> '5/13/2015 3:34:23 PM'
F13: 0705 -> '7/5/2015 12:00:00 AM'
F30: D00000000 -> 'D00000000'
F32: 0 -> '0'
F41: NOT DEFI -> 'NOT DEFI'
F43: BANK TO ZAP -> 'BANK TO ZAP '
F48: NAME -> 'NAME'
F49: 404 -> '404'
F102: 254734690270 -> '254734690270'(Source MSISDN)

F103: 254737561276 -> '254737561276'(Destination MSISDN)

Response to BANK

01931210F238000502A1800000000000060000001012345678902100000000000050000
5131429340411961429340513D0000000001000NOT DEFIBANK TO
ZAP 0004NAME4041225473469027012254737561276.

F2: 1234567890 -> '1234567890'
F3: 210000 -> '210000'
F4: 000000005000 -> '50.00'
F7: 0513142934 -> '5/13/2015 2:29:34 PM'
F11: 041196 -> '041196'
F12: 142934 -> '5/13/2015 2:29:34 PM'
F13: 0513 -> '5/13/2015 2:29:34 PM'
F30: D00000000 -> 'D00000000'
F32: 0 -> '0'
F39: 00 -> '00'
F41: NOT DEFI -> 'NOT DEFI'
F43: BANK TO ZAP-> 'BANK TO ZAP'
F48: NAME -> 'NAME'
F49: 404 -> '404'
F102: 254734690270 -> '254734690270'
F103: 254737561276 -> '254737561276'

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